To get the colour at pixel (x,y), use the following command:

```
color = `convert TMP2.mpc -channel rgba -alpha on -format
"%[pixel:u.p{x,y}]" info:`
```

If the colour is not 'none' but red, replace the contiguous red pixels with white, as follows:

```
convert TMP2.mpc -channel rgba -alpha on -fill white \
                    -draw "color x,y floodfill" TMP3.mpc
```

Now, you want only the white part. So fill all pixels that are not white with 'transparency', and then turn 'transparency' off so that all that is not white becomes black.

```
convert TMP3.mpc -channel rgba -alpha on -fill none +opaque
white -alpha off TMP3A.mpc
```

The white part is not a rectangle. So, clone the image and trim it so that the white part is bound by it. Now, replace all that is black with white in this trimmed image:

```
convert TMP3A.mpc -trim -fill white -opaque black TMP3B.mpc
```

Next, flatten the second image on top of the previous one to get the mask for a photo (see Figure 3).

```
convert TMP3A.mpc TMP3B.mpc -flatten TMP4.mpc
```

The above steps can be combined into a single *Convert* command as follows:

```
convert  \(TMP3.mpc -channel rgba -alpha on -fill none +opaque
white -alpha off  \) \
          \(+clone  -trim -fill white -opaque black \) \
          -flatten TMP4.mpc
```

The photo can now be extracted:

```
convert image.jpg tmp4.mpc -compose multiply -composite -trim
photo-1.jpg
```

While extracting it, you may want to add a step to straighten the image as well. So, instead, use the following command:

```
convert image.jpg tmp4.mpc -compose multiply -fuzz 6%
-composite -trim \
         -deskew 40% -trim +repage photo-1.jpg
```

The Multicrop script adds a border as well for a better presentation. Now, you need to remove the white image area so that it is not used again.

```
convert TMP3.mpc -channel
rgba -alpha on -fill none
-opaque white TMP2.mpc
```

Figure 1: The starting image

Figure 2: After removing the background and replacing the remaining image with red

Figure 3: Mask for extracting a photo

You are now ready to find another red pixel in *TMP2.mpc* and extract the next photo.

Usually, you may want to discard small photos as these may have spurious small islands of red. At times, you may find that the extracted image is smaller, e.g., if the sky is light, it may be mistaken for the background. So, there is considerable scope for making the script a lot smarter!

Tailpiece: Improving a scanned text page

Scanning a text page is never easy with Simple Scan, especially with old documents. Using the text mode, some folds show up as lines. If the text is faded or shaded, parts of the characters go missing. While the visual result of scanning in photo mode is much better, a printout normally has a distracting gray background and readability suffers in the process.

A solution is to use the 'white-threshold' option in Imagemagick's *Convert* command line utility after scanning a text document as a photo, as shown below:

```
$ convert scanned_text.jpg  -colorspace gray  -white-
threshold 60%  printable.jpg
```

END

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The author has earned the right to do what interests him. You can find him online at *http://sethanil.com* and *http://sethanil.blogspot.com*, and reach him via email at *anil@sethanil.com*